EXHAUST SHUTTER VALVE INSPECTION [SKYACTIV-D 1.5]

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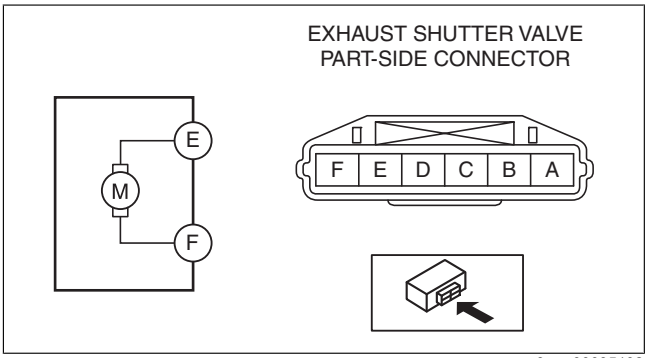
Note
- To determine the malfunctioning part, proceed with the diagnostics from “Function Inspection Using M-MDS”.
- The exhaust shutter valve cannot be removed because it is built into the middle pipe.

PID/DATA Monitor Inspection
1. Connect the M-MDS to the DLC-2.
2. Switch the ignition ON (engine off).
3. Display PID EXHSHUT_ACT and simulation item EXHSHUT_DSD using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)
4. Verify that the PID EXHSHUT_ACT value conforms to the simulation item EXHSHUT_DSD value when the exhaust shutter valve opening angle (simulation item: EXHSHUT_DSD) is changed using the simulation function.
   - If the PID EXHSHUT_ACT value does not conform at all or partially, perform the resistance inspection. (See Resistance Inspection.)
   - If the PID EXHSHUT_ACT value conforms, the exhaust shutter valve is normal.

Resistance Inspection
1. Switch the ignition OFF.
2. Disconnect the negative battery terminal. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
3. Disconnect the exhaust shutter valve connector. (See EXHAUST SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
4. Inspect the resistance between exhaust shutter valve terminals E and F.

   Exhaust shutter valve resistance (specification)
   0.3—100 ohms [20 °C {68 °F}]

   - If not within the specification, replace the exhaust shutter valve actuator. (See EXHAUST SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - If within the standard and the PID EXHSHUT_ACT value does not conform partially as a result of the PID/data monitor inspection, replace the middle pipe. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
   - If within the standard and the PID EXHSHUT_ACT value does not conform at all as a result of the PID/data monitor inspection, perform a visual inspection. (See Visual Inspection.)



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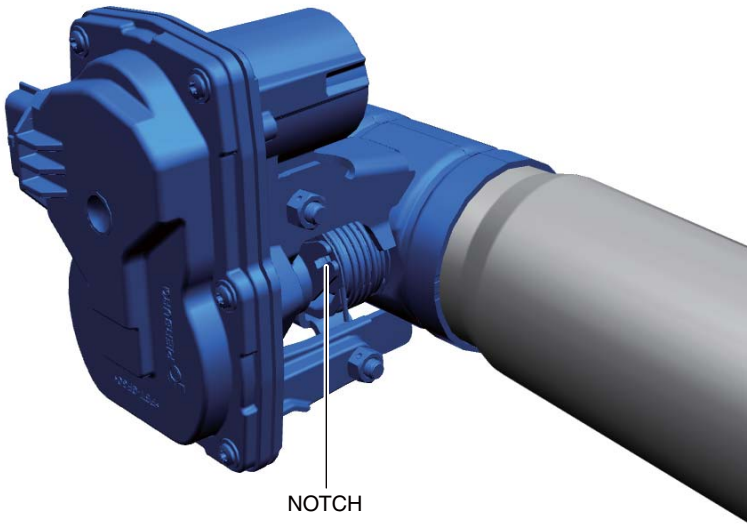
Visual Inspection
1. Connect the exhaust shutter valve connector. (See EXHAUST SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
2. Connect the negative battery terminal. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
3. Connect the M-MDS to the DLC-2.
4. Switch the ignition ON (engine off).
5. Display simulation item EXHSHUT_DSD using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].)

6. With the M-MDS connected to the DLC-2, move the vehicle to the position where the exhaust shutter valve motor can be verified as shown in the following figure.



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7. Verify that the notch of the exhaust shutter valve is driven as shown in the following figure when the exhaust shutter valve opening angle (simulation item: EXHSHUT_DSD) is changed using the simulation function.



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- If the notch does not operate, replace the middle pipe. (See EXHAUST SYSTEM REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
- If the notch operates, replace the exhaust shutter valve actuator. (See EXHAUST SHUTTER VALVE REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)